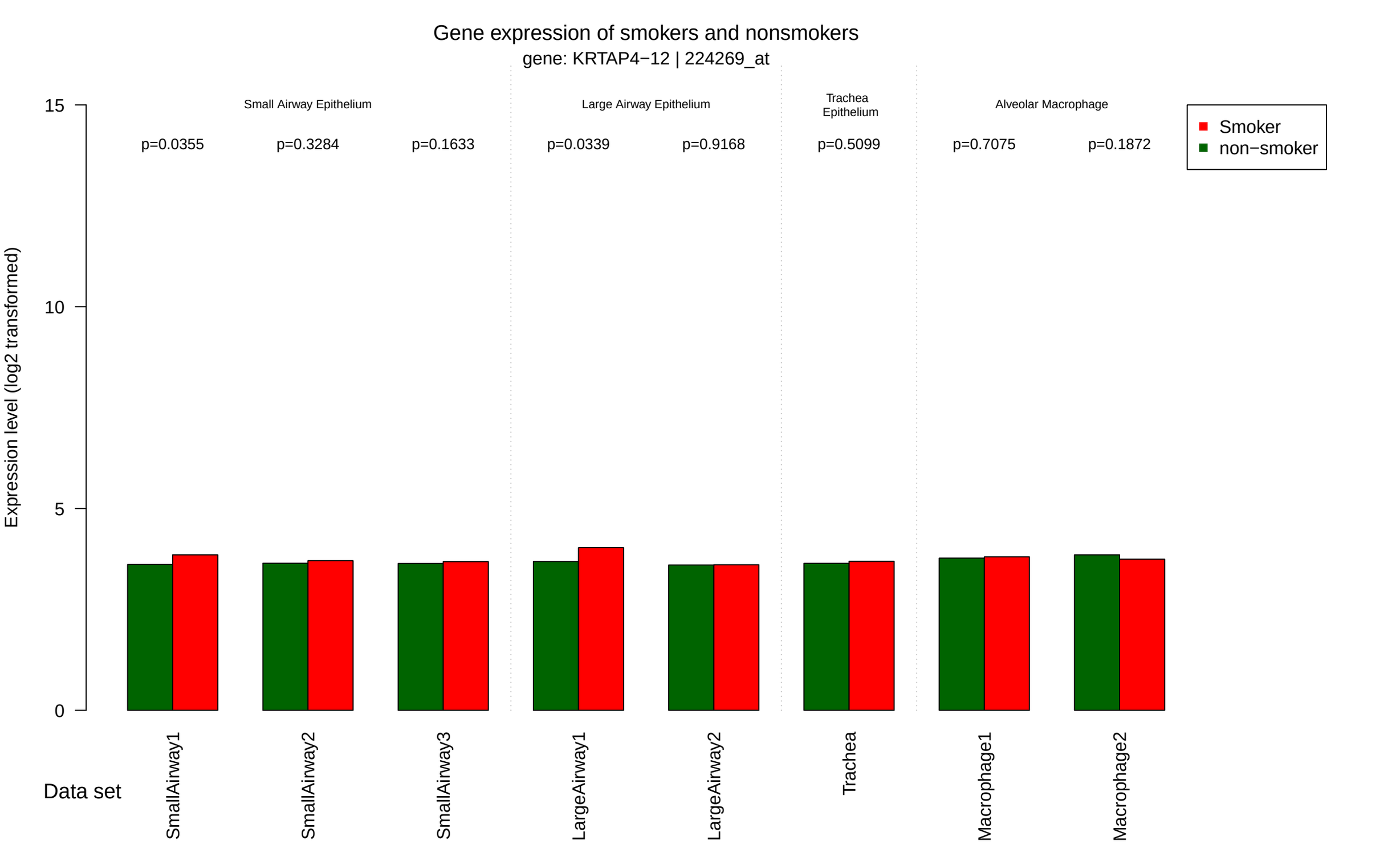
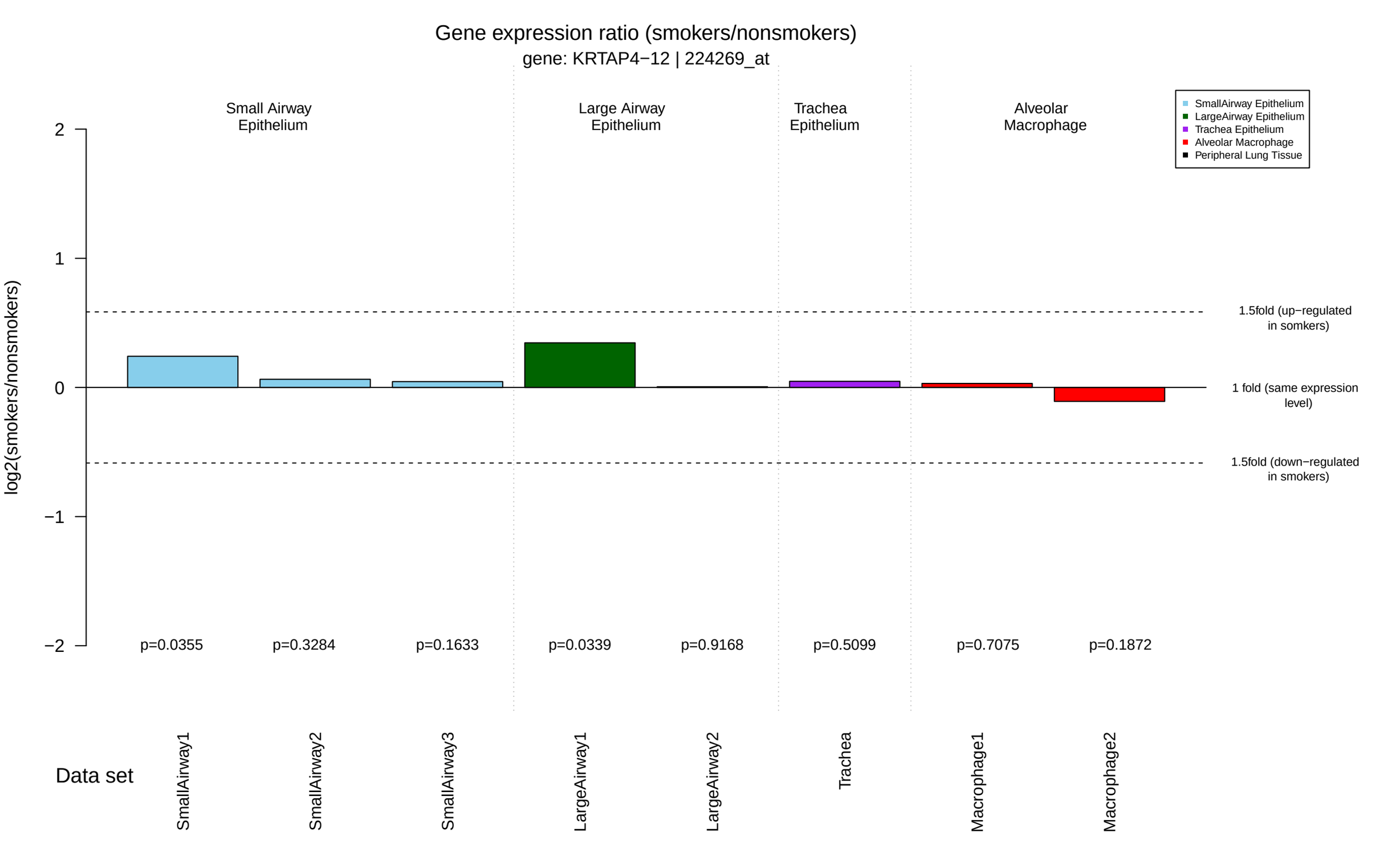
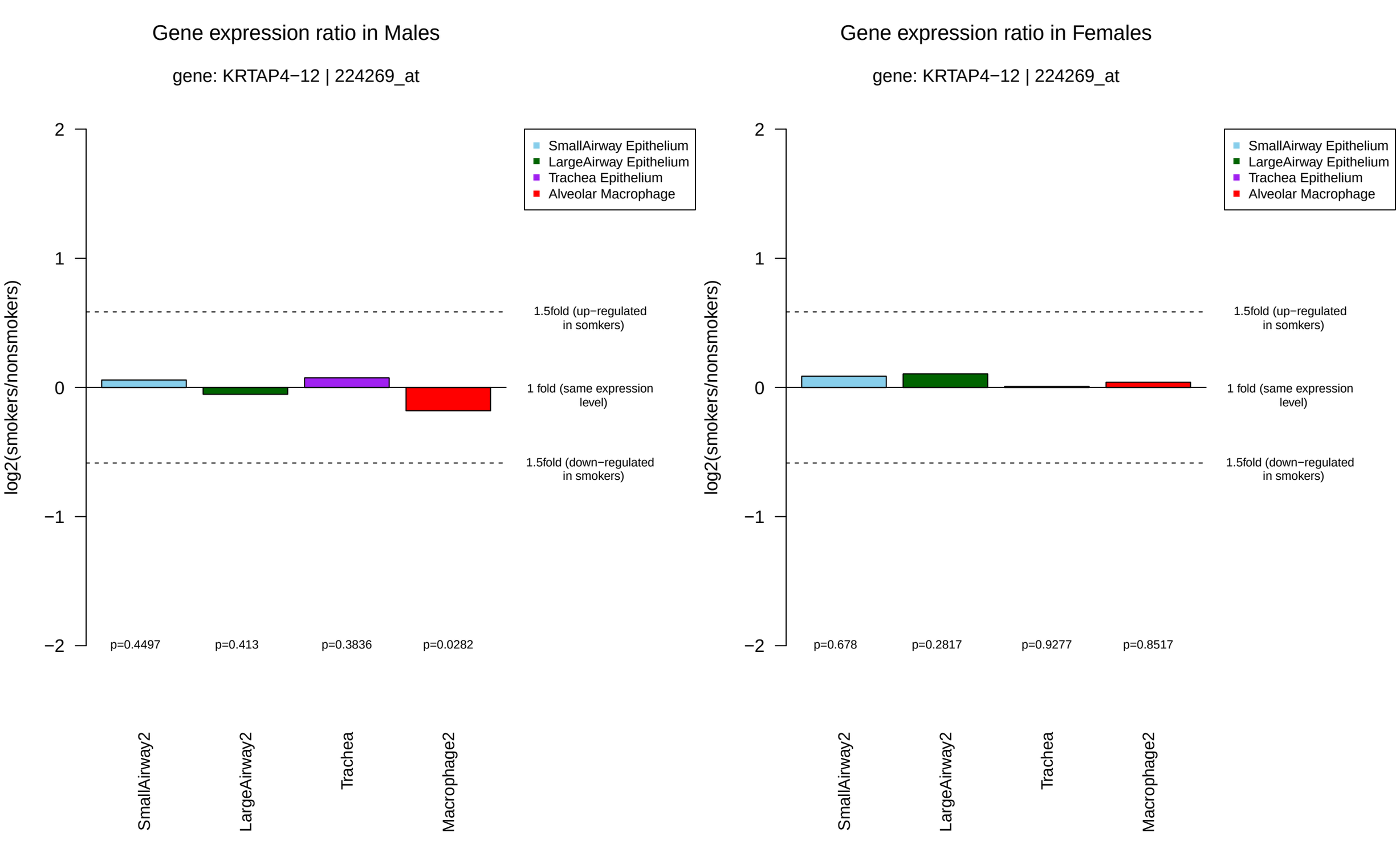
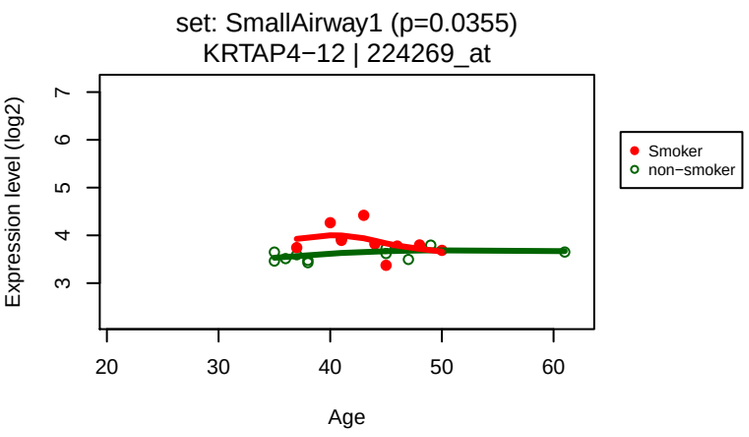
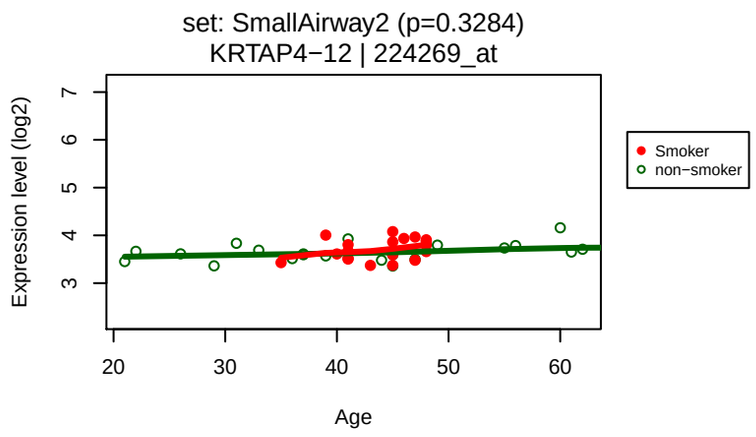
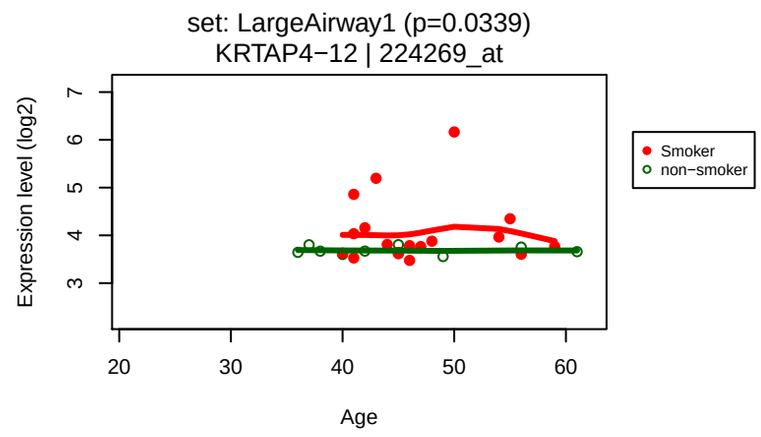

Gene expression of smokers and nonsmokers
gene: KRTAP4−12 | 224269_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.0355
p=0.3284
p=0.1633
p=0.0339
p=0.9168
p=0.5099
p=0.7075
p=0.1872
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

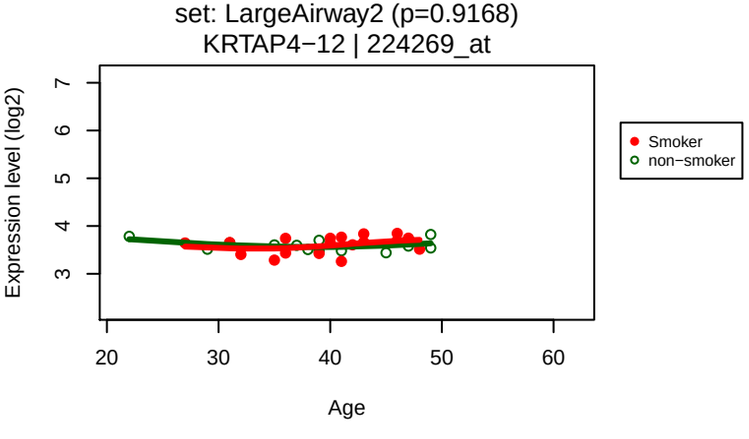

Gene expression ratio (smokers/nonsmokers)
gene: KRTAP4−12 | 224269_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0355
p=0.3284
p=0.1633
p=0.0339
p=0.9168
p=0.5099
p=0.7075
p=0.1872
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

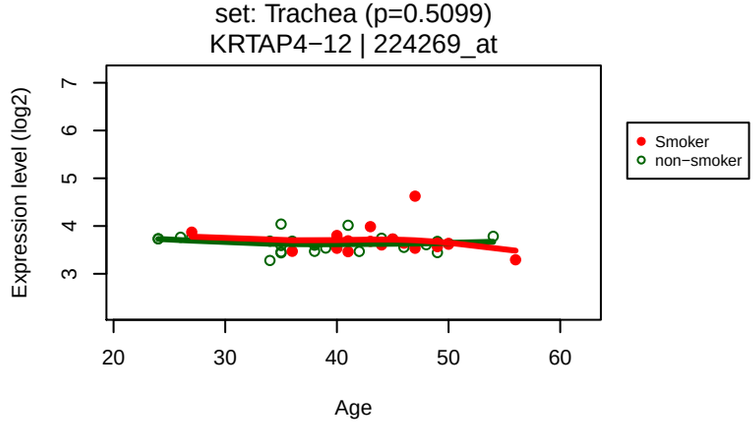

Gene expression ratio in Males
gene: KRTAP4−12 | 224269_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.4497
p=0.413
p=0.3836
p=0.0282
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: KRTAP4−12 | 224269_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.678
p=0.2817
p=0.9277
p=0.8517
SmallAirway2
LargeAirway2
Trachea
Macrophage2

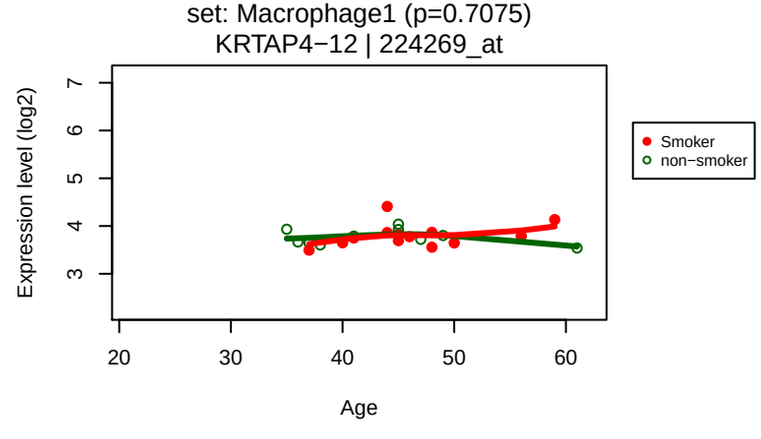

set: SmallAirway1 (p=0.0355)
KRTAP4−12 | 224269_at
Expression level (log2)
Age
Smoker
non−smoker

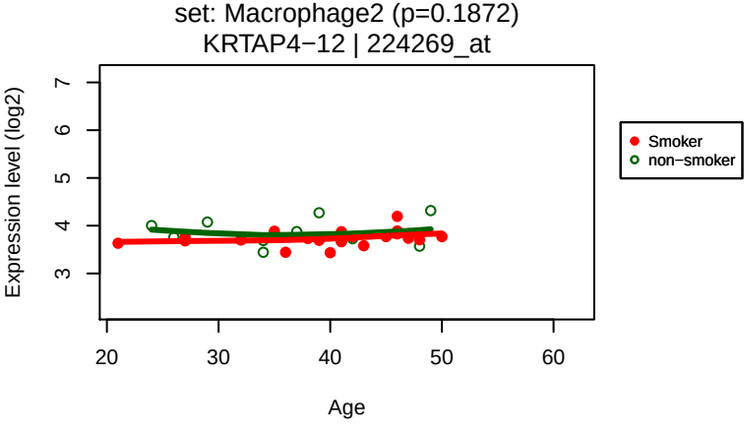

set: SmallAirway2 (p=0.3284)
KRTAP4−12 | 224269_at
Expression level (log2)
Age
Smoker
non−smoker

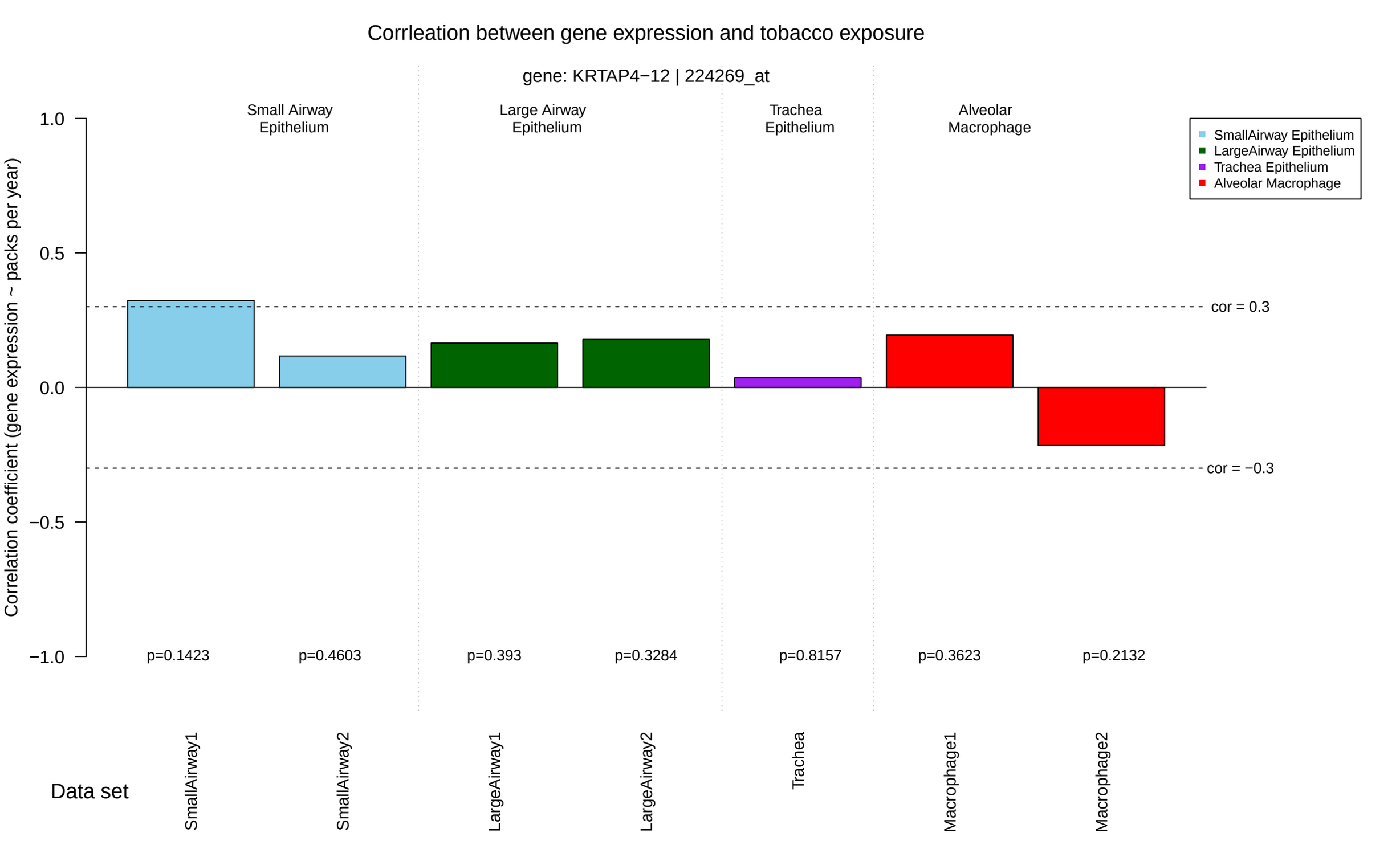

set: LargeAirway1 (p=0.0339)
KRTAP4−12 | 224269_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.9168)
KRTAP4−12 | 224269_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.5099)
KRTAP4−12 | 224269_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.7075)
KRTAP4−12 | 224269_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.1872)
KRTAP4−12 | 224269_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: KRTAP4−12 | 224269_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.1423
p=0.4603
p=0.393
p=0.3284
p=0.8157
p=0.3623
p=0.2132
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2